

Research Statement – Sean Patrick MacBride

My research focuses on exploring the cosmic frontier using astronomical instrumentation, multi-messenger astrophysics, and standard siren cosmology. In this proposal, I will describe how my past research experience and leadership has comprehensively prepared me to continue exploring the frontier as a **Lederman Fellow** at **Fermilab National Accelerator Laboratory**.

Early Career Contributions: My previous research has focused on instrumentation development for ground based optical telescopes, target-of-opportunity searches for electromagnetic counterparts of gravitational wave events, and constraining dark energy using electromagnetic and gravitational wave data.

Astronomical Instrumentation Development: My recent work has spanned both photometric and spectroscopic instruments. This began in 2020, with the **Large Lenslet Array Magellan Spectrograph (LLAMAS)**, where I validated the optical performance of the diffraction gratings and collimators for the LLAMAS spectrographs, assembled and verified optical tolerance for the 24 spectrograph cameras, and integrated all 2,400 AR-coated fibers into the focal plane with zero fiber loss, before it was commissioned on the Magellan Baade telescope in 2024 **??**. After integrating and testing LLAMAS, I shifted my focus from integral-field spectrographs to multi-object spectrographs. Starting in 2023, while at the University of Michigan-Ann Arbor, I tested a common failure mode on **Dark Energy Spectroscopic Instrument (DESI)** robotic positioners, where I found that re-orienting the robotic positioner to an inverted physical configuration could eliminate the failure mode. This indicated that gravity-sag has a significant influence on the presence and recovery of these failures **(?)**. This expertise with spectroscopic instruments translated effectively to the transcendent photometric imager of the next decade, the **LSST Camera (LSSTCam)**. Shortly after beginning my PhD at the University of Zürich, I moved to Chile in 2024 to serve as an LSSTCam Summit Support Scientist, where **I led LSSTCam focal-plane commissioning** on Cerro Pachón as a principal component of the in-kind contribution from Switzerland to the Rubin Project. In this role, I tested the CCDs in both the summit clean-room and on the Simonyi Survey Telescope, and optimized the sensor performance. This characterization and testing played a leading role in achieving LSSTCam first photon in April 2024 (see Figure 1a). This work culminated in an invited proceeding to the SPIE Astronomical Telescopes + Instrumentation conference in July 2026, an impactful astronomical instrumentation conference held biennially **(?)**.

Target-of-Opportunity Searches: Since 2023, I have led twenty observing campaigns using different telescopes and facilities, with the goal of detecting electromagnetic (EM) counterparts to gravitational wave (GW) merger events. A variety of multi-messenger analyses are enabled by the detection of an EM counterpart to a GW merger event, and my particular focus is to combine the redshift information from the host galaxy of an EM

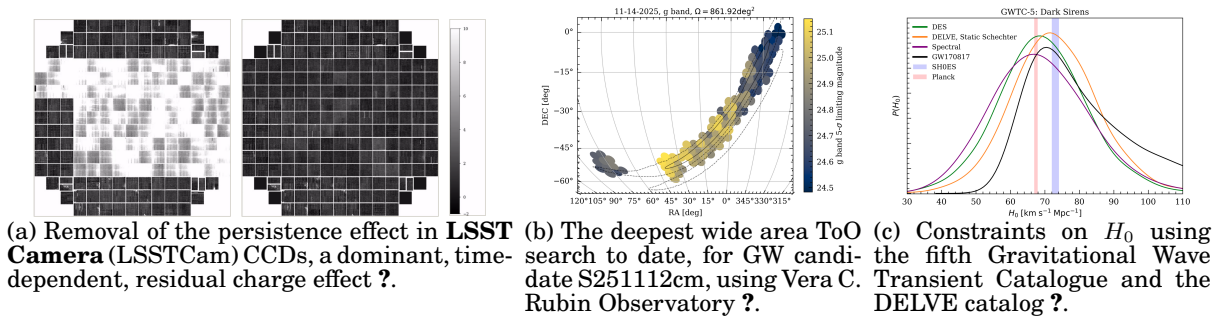


Figure 1: A summary of early career contributions, spanning instrumentation, multi-messenger astrophysics, and standard siren cosmology.

counterpart with the luminosity distance inferred from the GW source. With this combination, it is possible to constrain the local rate of accelerated expansion of the universe, known as the Hubble Constant (H_0) in the standard model of cosmology, Λ CDM. This technique is known as the *bright siren* method ?.

Since 2023, I have been co-leading the Dark Energy Survey Gravitational Wave follow-up program (DESGW) program, conducting **Dark Energy Camera** (DECam) observations, processing raw image data from sixteen Target-of-Opportunity (ToO) campaigns to calibrated products, and analyzing processed data in response to LIGO-Virgo-Kagra (LVK) Observing Run 4 (O4) triggers. During a recent observing campaign in 2025, I led a collaboration between DESGW and the Japanese collaboration for gravitational-wave electromagnetic follow-up (J-GEM) when a well-localized gravitational wave merger candidate was accessible to both Northern and Southern hemisphere instruments. In this collaboration, I obtained multi-epoch photometry using DECam, processed the obtained data, and cross-matched this data with spectroscopic observations using **Prime Focus Spectrograph** (PFS), a recently commissioned multi-object spectrograph at the Subaru Telescope. This analysis enabled the first scientific result using PFS (?). This data serves a dual purpose, providing valuable transient characterization information, and also serving as the foundation for a measurement of H_0 using the dark siren method with spectroscopic data (see Figure 2).

Since 2025, I have been leading the Vera C. Rubin Observatory (Rubin) ToO observing program as the Rubin ToO observer. In this role, I have led observing responses to one interstellar object (?), one high-energy neutrino event (later reclassified as terrestrial), and two gravitational-wave merger candidates, including the deepest, wide-area GW ToO search to date ($m_{g,5\sigma} \sim 24.8$ across $\sim 850 \text{ deg}^2$, see Figure 1b) (??). The process to build, test, and deploy an automated ToO system for Rubin is reflected in the recent proceedings of SPIE Astronomical Telescopes + Instrumentation 2026 (?), closely following the community-defined science case for the program (?).

In addition to building the Rubin ToO system, I have led extensive testing of the Rubin ToO infrastructure with the LVK and IceCube Neutrino Observatory collaborations, including two mock-data-challenges to assess the average latency of a Rubin ToO response, the accuracy of our alert processing pipeline, and the sensitivity of simulated Rubin observations to optical counterparts (?). **Add a metric from the mock-data-challenge.**

TODO

Standard Siren Constraints on the Hubble Constant: While recent observing campaigns have not yet yielded an EM counterpart to a GW merger, I have retained a focus on constraining cosmology using EM and GW data, employing the *dark siren* method to measure the local expansion rate of the universe, H_0 . The dark sirens method infers redshift information of GW events using a catalog of potential host galaxies, which is lower precision than the bright siren method, but can be applied to all GW events ?. My work began by supporting the preparation of Dark Energy Survey (DES) data, resulting in the first dark siren constraint using a deep photometric catalog to jointly infer GW population parameters and cosmological parameters (?). DES was accepted as the fiducial result for the most recent LVK analysis, and demonstrated a 22% improvement in constraining H_0 compared to previous analyses (?). I have since expanded on the DES preparation, leading a dark-siren analysis using Dark Energy Local Volume Exploration (DELVE), where initial results show that when photometric redshift biases are properly treated, **DELVE outperforms DES using dark sirens** (see Figure 1c), constraining H_0 to 11% when combined with the bright siren GW170817 (?).

Future Research Activities: My proposed research builds on this foundation through complementary directions. First, I will characterize image sensors for future cosmic surveys, while also deploying monitoring tools for the LSST Camera to improve LSST data quality. Second, I will lead Target-of-Opportunity searches to detect the next bright siren

and other exotic astrophysical events. Third, I will usher standard siren cosmology into a precision era, leveraging LSST data alongside specialized observational programs for bright and dark siren cosmology.

Astronomical Instruments to Leverage LSST data: In the Legacy Survey of Space and Time (LSST) era, the central bottleneck will not be imaging depth, but rather spectroscopic capacity. To aid in this effort, I will test and deploy advanced focal plane technology for future multi-object spectrographs. As part of this focus, I will explore instrument design for a multi-object spectrograph optimized for standard siren cosmology, specialized for the precision and event rates of forthcoming gravitational wave detectors. This instrument would serve two goals: 1) characterize astrophysical transients associated with bright standard sirens, and 2) obtain spectroscopic redshifts from host galaxies to bolster dark siren measurements. Through this dual purpose, the proposed instrument would become a dark energy science machine, primed to capitalize on the data deluge from LSST and future GW observatories. These pathfinding studies will inform the proposed **Stage-5 Spectroscopic Experiment** (Spec-S5), a future, large-scale cosmic frontier experiment planned for the mid-2030's ?. Developing focal-plane technology for other instruments will help shape the capabilities of Spec-S5, so it serves as an effective spectroscopic complement to Rubin.

In addition to testing image sensors for future astronomical instruments, I will sustain a support-scientist role for LSSTCam as the LSST transitions into survey operations, maintaining the operational knowledge needed to respond quickly to sensor anomalies and focal-plane issues as they arise. Building on my commissioning-era characterization of LSSTCam defects, I will also deploy a defect-monitoring tool, currently in-development, to `cp_verify`, the calibration-product verification framework used by the LSST Science Pipelines ?. This tool will elevate defect monitoring to an automated process, improving LSST data quality.

Bright Siren Searches Enabled by Rubin Observatory: I will lead the observing planning, execution, and analysis of the next bright standard siren accessible to Rubin Observatory,

along with other ToO's pursued during the LSST, continuing my role as the Rubin ToO observer throughout the transition to operations. This leadership is enabled by my specialized expertise in the alert processing of the ToO system, my intimate knowledge of the technical strengths and limitations of the observatory, and my sustained development of the ToO system within the wider Rubin ecosystem. While it is difficult to predict the timing of the next bright siren, I will continue to deploy improvements to the ToO system to improve response latency, observing strategy, and automation, aiming to maximize the forthcoming opportunity to detect and characterize these optical counterparts to obtain a well-constrained measure of H_0 .

In parallel to operating the Rubin ToO system, I will analyze DESGW candidates identified during future LVK observing runs. This positions me to sustain and support Rubin's ToO program with additional imaging in other filters and extend the temporal-sampling

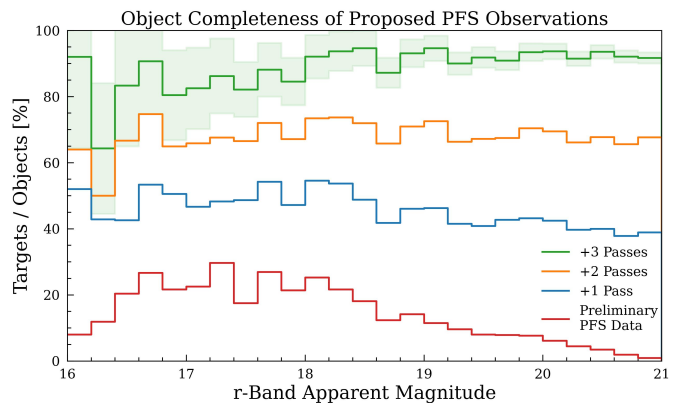


Figure 2: In red, the completeness of PFS objects acquired as part of our ToO campaign ?, with other colors owing the forecasted completeness of proposed observations. The high completeness obtained over the localization region in 3 passes (one night) demonstrates the utility of current and future multi-object spectrographs as characterization machines for standard siren cosmology.

using a similarly capable photometric imager. This complimentary support will play an important role, especially as other priorities of the LSST are balanced against the ToO program. As the LSST ToO program matures, I will work to enhance transient characterization by securing spectroscopic follow-up time on other instruments, moving beyond photometric classification to confirm spectral features that can inform the scope of Rubin's ToOs, and constrain the physical properties of LSST transients.

This includes an ongoing program with J-GEM using PFS, a new program using LLMAS to characterize promising candidates from the LSST alert stream, and a future program using data from the BOOMBOX spectrograph from the Via project. **Double check if applying to a Carnegie institution, otherwise swap with SOAR. TODO**

Looking further ahead, I will begin preparing Rubin's ToO infrastructure for multi-messenger astrophysics of the future, including third generation (3G) ground-based gravitational-wave detectors, the space-based GW detector Laser Interferometer Space Antenna (LISA), new classes of high-energy neutrinos from IceCube gen-2, and the forthcoming long-baseline neutrino experiments, Deep Underground Neutrino Experiment (DUNE) and Hyper Kamiokande (Hyper-K). This era will usher in larger event rates, more precise localizations, and new source classes (including massive black hole mergers), that demand ToO strategies tailored for a new status quo. This work will focus on identifying the optimal observing strategies for Rubin discovery using `rubin_sim` simulations along with simulated light curves, updating the alert-handling infrastructure for rapid processing, and strengthening follow-up partnerships across a variety of other observatories and experiments. Enhancing the detection of counterparts associated with GWs from 3G and space-based detectors will extend the standard siren lever arm for precision cosmology, enabling measurements of Ω_m , w , and other dark energy models.

Dark Siren Measurements for Precision Cosmology: I will pursue a cutting-edge program that creates a gold-standard photometric galaxy catalog for dark siren cosmology using LSST data. This program is structured in four stages, each building towards a powerful, world leading measurement. It begins with an ongoing project, focused on characterizing the LSST Data Preview 2 (DP2) galaxy catalog, aiming to define robust and scalable object classification for future LSST data releases. Following characterization of DP2, I will finish a forecasting study that quantifies the H_0 precision achievable by combining simulated first-year LSST data with an O4-like gravitational-wave catalog, extending the dark-siren methodology from my recent DELVE measurement to project the constraining power in the LSST era. Superseding DP2 preparation, I will characterize the first LSST Data Release (DR1) galaxy catalog and produce a dark siren measurement of H_0 . This result will extend the methodology I developed with DELVE to a dataset with substantially greater depth, area, and photometric uniformity, all features that will make it the standard galaxy catalog for dark siren analyses. Building on the DR1 measurement, I aim to combine LSST large-scale structure measurements in a joint cosmological analysis, using cosmic shear and standard sirens to obtain a tighter constraint of H_0 , and extend the constraints to Ω_m .

As the wealth of GW data increases alongside a wide variety of new galaxy catalog data, I will explore the frontier of the standard siren method using specialized surveys and forthcoming data releases from other surveys, while engaging with the Horizon Europe-funded GWCats collaboration, a tight-knit group of researchers focused on using galaxy catalogs and GW data to constrain cosmology. To assess the frontier of the dark siren method using observations, I will lead a dedicated spectroscopic dark-siren pilot survey, obtaining targeted redshifts within GW localization volumes to assess the limits and enhancements to H_0 constraints using well-localized dark sirens. This survey is led through a continuing collaboration with J-GEM on the PFS instrument, obtaining additional observations for the GW merger pursued in our collaborative DESGW ToO with an expected observation

date starting in April 2027 (see Figure 2). In parallel, I will lead the image processing for a similar survey, the Gravitational Wave Photometric Redshift Improvement with Medium-band Observations (GW-Primo) project, to test how medium band observations improve H_0 precision through enhanced photometric redshifts. By comparing the results of these two survey programs, I will be able to assess which route, enhanced photometric or spectroscopic, will provide the best constraints on H_0 given limited telescope time and the current sample of well-localized GWs.

Similar to DES the The Ultraviolet Near-infrared Optical Northern Survey (UNIONS) survey provides a large area galaxy catalog in the Northern hemisphere, where DES and LSST coverage is not possible. I will prepare the UNIONS galaxy catalog for dark siren cosmology through ongoing work with the UNIONS collaboration, applying the lessons from the DELVE galaxy catalog preparation. I will construct both an isolated UNIONS catalog and a combined galaxy catalog of UNIONS and LSST data for standard siren cosmology, while taking care to control any systematics that come from joining different galaxy catalogs. This combined catalog will span most of the accessible extragalactic sky, including several well localized GWs in the northern hemisphere, becoming a value-added catalog compared to LSST data alone.

Finally, with the new discovery capabilities of Rubin for bright standard sirens, the systematics of multiple bright siren measurements must be thoroughly examined before claiming a robust and precise measurement of H_0 . To build this foundation, I will lead an assessment of peculiar velocity systematics in bright siren measurements, using synthetic and real data to accurately map the velocity field and reduce the systematic contribution to future bright siren measurements.

As a **Lederman Fellow**, I will achieve these research goals by collaborating with **Juan Estrada and Alex Drlica-Wagner** to **effectively characterize image sensors for future spectroscopic surveys**, in addition to working with **Ken Herner** to **search for the next electromagnetic counterpart of a gravitational wave event using Rubin Observatory**. Through this concerted effort, I will propel advancements in astronomical instrumentation, push the boundaries of multi-messenger astrophysics, and mature standard sirens, both bright and dark, into a precision cosmological probe.

References